Case Study: Retail Supply Chain Optimization

Industry: Grocery Retail | Engagement Length: 7 months | Practice: Operations

Client Background

The client is a mid-sized regional grocery retailer operating 45 stores across three states, with an annual revenue of approximately $310 million. The company had grown through acquisition over the prior decade, resulting in inconsistent inventory practices between store clusters that had originally belonged to different regional chains.

Challenge

The client's inventory forecasting relied on spreadsheets updated manually by each store manager, with no shared visibility across locations. This led to frequent stockouts of high-demand items during weekend peaks and significant overstock of perishable goods early in the week, resulting in an estimated $2.3 million in annual waste, primarily in the produce and dairy categories.

A secondary challenge was staff time: store managers reported spending eight to ten hours per week on manual counts and reorder spreadsheets, time that could otherwise go toward customer-facing activities during a period when the client was also trying to improve in-store service scores.

Approach

Our team implemented a centralized demand forecasting system that ingests point-of-sale data nightly and generates automated purchase order recommendations for each store location, factoring in day-of-week seasonality, local weather forecasts, and historical promotional lift. We also introduced a weekly variance review process so store managers could flag local anomalies, such as a nearby competitor closing or a local event driving unusual foot traffic.

The rollout was phased across three regions over four months, starting with the five highest-volume stores to validate the forecasting model before expanding company-wide. Store managers received two half-day training sessions, and a dedicated support channel was staffed for the first six weeks after each region's go-live to handle edge cases the model had not yet learned.

A key design decision was to keep store managers in the loop rather than fully automating purchase orders. The system generates a recommendation, but managers retain override authority, which significantly improved adoption compared to a fully automated pilot the client had attempted internally two years earlier.

Results

Within six months of full deployment, the client reported a 34% reduction in perishable waste and a 19% reduction in stockout incidents on top-selling SKUs. Store managers reported spending roughly six fewer hours per week on manual inventory tasks, time that store-level leadership redirected toward staff coaching and customer service initiatives.

Override rates on system-generated purchase orders dropped from 40% in the first month to under 12% by month five, indicating growing trust in the model's recommendations. The client's finance team estimated the engagement paid for itself within the first eight months through waste reduction

alone, before accounting for labor savings or reduced stockout-driven lost sales.

Next Steps

The client has since asked us to extend the same forecasting approach to their newly acquired distribution warehouse, and is evaluating whether to apply a similar variance-review model to non-perishable categories, where waste is lower but carrying costs on overstocked slow-moving items remain a concern for the finance team.